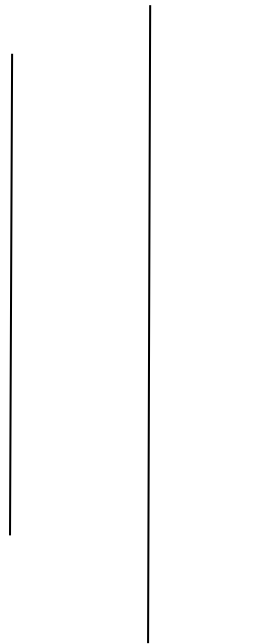




**Tribhuvan University
Institute of Science & Technology
Samriddhi College**



A
Lab Report
on
Advanced Java Programming
[Subject Code: CSC409]

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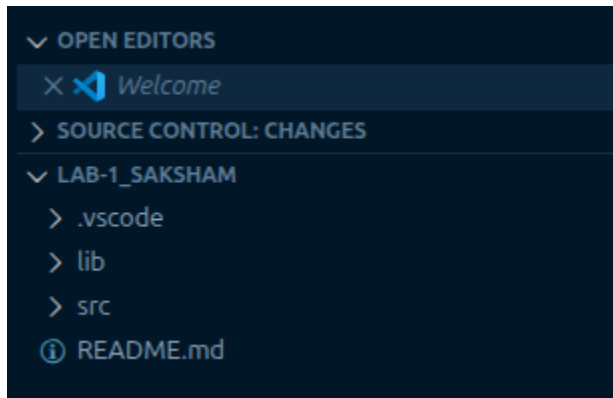
List of Programs:

S.No.	Name of the Program	Page No.	Date	Sign
1.	Write a GUI program using components to find the sum and difference of two numbers. Use two text fields for giving input and a label for output. The program should display the sum if the user presses the mouse and the difference if the user releases the mouse.			
2.	Write a Program using swing components to add two digits. Use text fields for inputs and output. Your program should display the result when the user presses a button.			
3.	Write a program to Create a Form Below using Swing and perform Add, Subtract, Multiply and Divide as user click button with selected option.			
4.	Write a GUI program using components to find the sum and difference of two numbers. Use two text fields for giving input and a label for output. The program should display the sum if the user presses the mouse and the difference if the user releases the mouse using adapter class.			
5.	Create a Simple Login Form using GridLayout in Java.			
6.	Create a Simple Login Form using GridBagLayout in Java.			
7.	Write a program that divides the frame into five regions by using border layout and then add panels in the east, north and center region. Finally add some descriptive label in the north panel, buttons with icon in the east panel and a sample form in the center panel. You can further subdivide the center panel, if necessary. Prepare a program with three text boxes First Number, Second Number, and Result and four buttons add, subtract, multiply and divide. Handle the events to perform the required operation and display results.			
8.	Design a form with three buttons with captions "RED," "BLUE," and "GREEN," respectively. Then write a program to handle the event such that when the user clicks the button, the color of that button will be the same as its caption.			
9.	Write a Java program using Swing that demonstrates the use of different layout managers (FlowLayout, GridLayout, and BorderLayout). • Divide the main frame into three panels using BorderLayout: - o North Panel: Use FlowLayout to arrange three labels (Name, Roll No, Class) in a single row. - o Center Panel: Use GridLayout (3x2) to create a simple login form with two text fields (Username, Password), a JCheckBox ("Remember Me"), and a JButton ("Login").			

	- o South Panel: Use <code>FlowLayout</code> to add two buttons (<code>Reset</code> and <code>Exit</code>). • Handle events so that: - o When Login is pressed, display the entered username in a message dialog. - o When Reset is pressed, clear the text fields. - o When Exit is pressed, close the program.			
10.	Design a form with three buttons with captions “Circle,” “Square,” and “Triangle.” • When the user clicks the “Circle” button, draw a red circle in the center of the frame. • When the user clicks the “Square” button, draw a blue square in the center of the frame. • When the user clicks the “Triangle” button, draw a green triangle in the center of the frame.			
11.	Design a Swing application that contains three radio buttons labeled "Red", "Green", and "Blue". When the user selects a radio button, the background color of the window should change to the selected color. Implement event handling using an <code>ItemListener</code>.			
12.	Write a Java Swing program to create a toolbar with three buttons. Each button should display an icon (New, Login, and Logout). Place the toolbar at the top of the window using a suitable layout. When a button is clicked, display a message dialog showing the name of the action (e.g., “New File Created”, “Login Successful”, “Logged Out”).			

LAB 1 - Write a GUI program using components to find the sum and difference of two numbers. Use two text fields for giving input and a label for output. The program should display the sum if the user presses the mouse and the difference if the user releases the mouse.

SOLUTION EXPLORER:



Program:

```
import javax.swing.*.*;
import java.awt.*.*;
import java.awt.event.*;

public class SumDifferenceSwing {

    public static void main(String[] args) {

        // Create Frame
        JFrame frame = new JFrame("Sum and Difference");
        frame.setSize(400, 200);
        frame.setLayout(new FlowLayout());
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        // Create Components
        JTextField textField1 = new JTextField(10);
        JTextField textField2 = new JTextField(10);
        JLabel resultLabel = new JLabel("Result will appear here");

        // Add Components to Frame
        frame.add(new JLabel("Enter First Number:"));
        frame.add(textField1);

        frame.add(new JLabel("Enter Second Number:"));
        frame.add(textField2);

        frame.add(resultLabel);

        // Add Mouse Listener to Frame
        frame.addMouseListener(new MouseAdapter() {
```

```

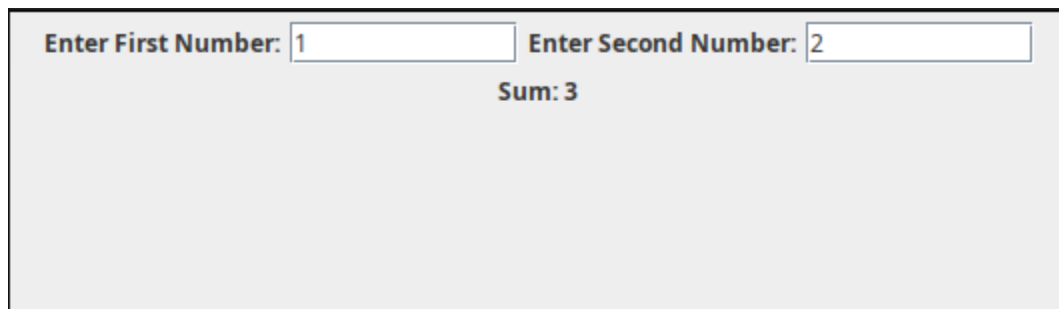
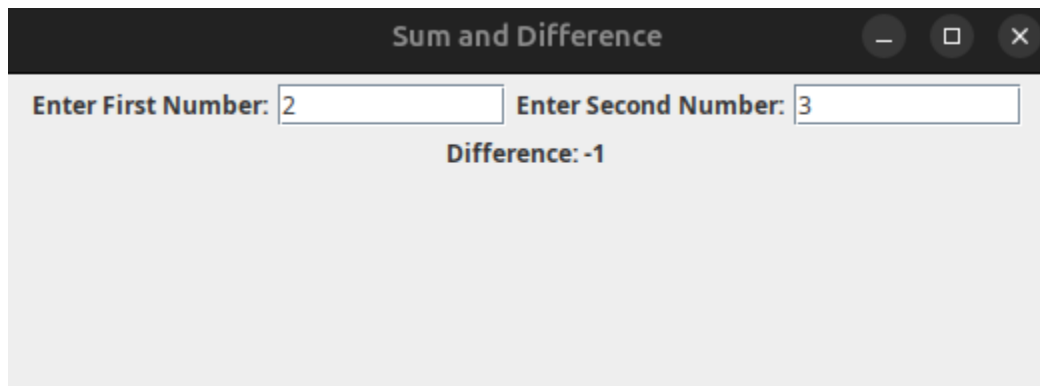
public void mousePressed(MouseEvent e) {
    try {
        int num1 = Integer.parseInt(textField1.getText());
        int num2 = Integer.parseInt(textField2.getText());
        int sum = num1 + num2;
        resultLabel.setText("Sum: " + sum);
    } catch (Exception ex) {
        resultLabel.setText("Enter valid numbers");
    }
}

public void mouseReleased(MouseEvent e) {
    try {
        int num1 = Integer.parseInt(textField1.getText());
        int num2 = Integer.parseInt(textField2.getText());
        int diff = num1 - num2;
        resultLabel.setText("Difference: " + diff);
    } catch (Exception ex) {
        resultLabel.setText("Enter valid numbers");
    }
}
});

frame.setVisible(true);
}
}

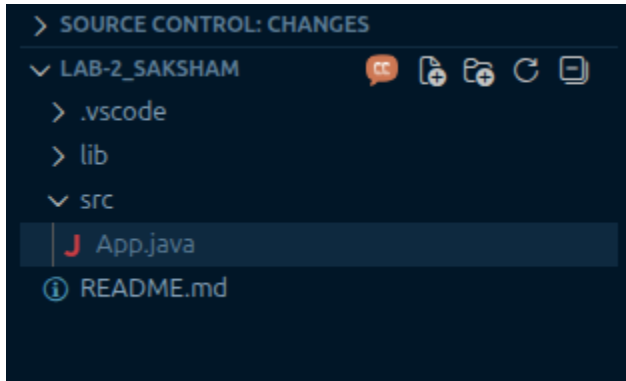
```

OUTPUT



LAB2 - Write a Program using swing components to add two digits. Use text fields for inputs and output. Your program should display the result when the user presses a button.

SOLUTION EXPLORER:



PROGRAM

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class Lab2AddDigits {

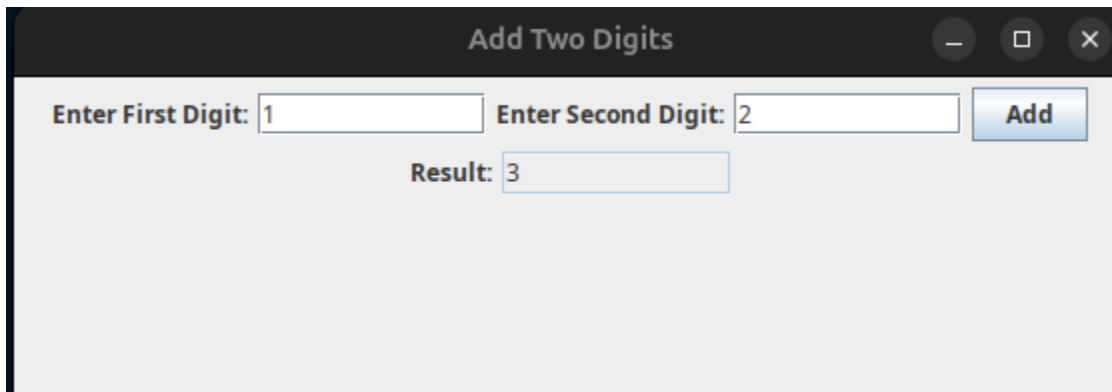
    public static void main(String[] args) {

        // Create Frame
        JFrame frame = new JFrame("Add Two Digits");
        frame.setSize(400, 200);
        frame.setLayout(new FlowLayout());
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        JTextField textField1 = new JTextField(10);
        JTextField textField2 = new JTextField(10);
        JTextField resultField = new JTextField(10);
        resultField.setEditable(false); // Output only
        JButton addButton = new JButton("Add");
        frame.add(new JLabel("Enter First Digit:"));
        frame.add(textField1);
        frame.add(new JLabel("Enter Second Digit:"));
        frame.add(textField2);
        frame.add(addButton);
        frame.add(new JLabel("Result:"));
        frame.add(resultField);
        addButton.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                try {
                    int num1 = Integer.parseInt(textField1.getText());
                    int num2 = Integer.parseInt(textField2.getText());
                    int sum = num1 + num2;
                    resultField.setText(String.valueOf(sum));
                } catch (NumberFormatException ex) {
                    JOptionPane.showMessageDialog(frame, "Invalid input. Please enter numbers only.", "Error", JOptionPane.ERROR_MESSAGE);
                }
            }
        });
    }
}
```

```
        } catch (NumberFormatException ex) {  
            resultField.setText("Invalid Input");  
        }  
    }  
});  
  
frame.setVisible(true);  
}  
}
```

Output:

```
• sakshamrimal@sakshamrimal-Latitude-5430:~/java/lab/Lab-2_Saksham$ cd src  
• sakshamrimal@sakshamrimal-Latitude-5430:~/java/lab/Lab-2_Saksham/src$ javac Lab2AddDigits.java  
• sakshamrimal@sakshamrimal-Latitude-5430:~/java/lab/Lab-2_Saksham/src$ java Lab2AddDigits  
█
```



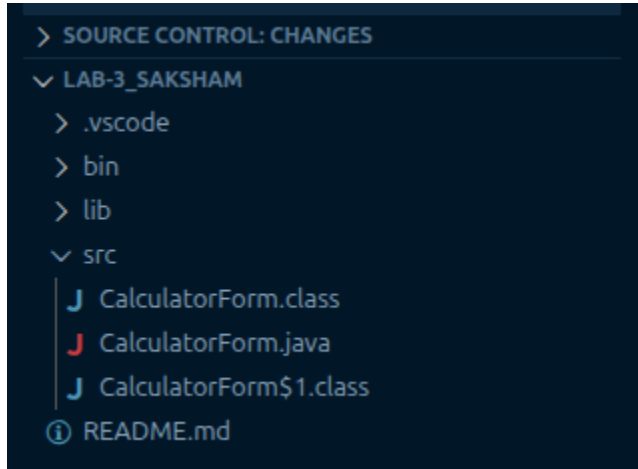
Add Two Digits

Enter First Digit: 1 Enter Second Digit: 2 Add

Result: 3

LAB 3: Write a program to Create a Form Below using Swing and perform Add, Subtract, Multiply and Divide as user click button with selected option.

SOLUTION EXPLORER:



PROGRAM:

```
import javax.swing.*.*;
import java.awt.*.*;
import java.awt.event.*;

public class CalculatorForm {
    public static void main(String[] args) {
        JFrame frame = new JFrame("Simple Calculator Form");
        frame.setSize(400, 300);
        frame.setLayout(new GridLayout(6, 2, 10, 10));
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        JLabel label1 = new JLabel("Enter First Number:");
        JTextField textField1 = new JTextField();

        JLabel label2 = new JLabel("Enter Second Number:");
        JTextField textField2 = new JTextField();

        JLabel resultLabel = new JLabel("Result:");
        JTextField resultField = new JTextField();
        resultField.setEditable(false);

        JButton addButton = new JButton("Add");
        JButton subButton = new JButton("Subtract");
        JButton mulButton = new JButton("Multiply");
        JButton divButton = new JButton("Divide");
        frame.add(label1);
        frame.add(textField1);
```

```

frame.add(label2);
frame.add(textField2);

frame.add(addButton);
frame.add(subButton);

frame.add(mulButton);
frame.add(divButton);

frame.add(resultLabel);
frame.add(resultField);
ActionListener action = new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        try {
            double num1 = Double.parseDouble(textField1.getText());
            double num2 = Double.parseDouble(textField2.getText());
            double result = 0;
            if (e.getSource() == addButton)
                result = num1 + num2;
            else if (e.getSource() == subButton)
                result = num1 - num2;
            else if (e.getSource() == mulButton)
                result = num1 * num2;

            else if (e.getSource() == divButton) {
                if (num2 == 0) {
                    resultField.setText("Cannot divide by zero");
                    return;
                }
                result = num1 / num2;
            }
            resultField.setText(String.valueOf(result));
        } catch (NumberFormatException ex) {
            resultField.setText("Invalid Input");
        }
    }
};
addButton.addActionListener(action);
subButton.addActionListener(action);
mulButton.addActionListener(action);
divButton.addActionListener(action);

frame.setVisible(true);
}
}

```

Output:

Simple Calculator Form	
Enter First Number:	2
Enter Second Number:	4
Add	Subtract
Multiply	Divide
Result:	6.0

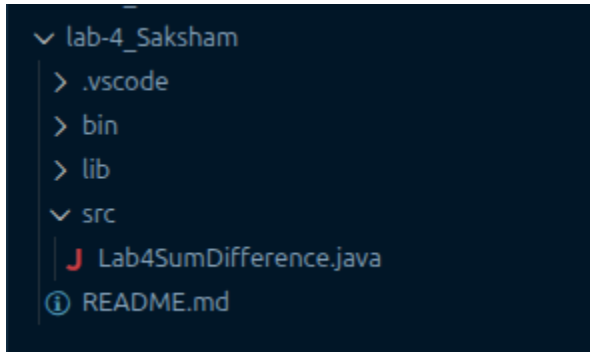
Simple Calculator Form	
Enter First Number:	2
Enter Second Number:	4
Add	Subtract
Multiply	Divide
Result:	-2.0

Simple Calculator Form	
Enter First Number:	2
Enter Second Number:	4
Add	Subtract
Multiply	Divide
Result:	8.0

Simple Calculator Form	
Enter First Number:	2
Enter Second Number:	4
Add	Subtract
Multiply	Divide
Result:	0.5

LAB-4 : Write a GUI program using components to find the sum and difference of two numbers. Use two text fields for giving input and a label for output. The program should display the sum if the user presses the mouse and the difference if the user releases the mouse using adapter class.

SOLUTION EXPLORER:



PROGRAM

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class Lab4SumDifference {

    public static void main(String[] args) {

        JFrame frame = new JFrame("LAB-4: Sum and Difference");
        frame.setSize(400, 200);
        frame.setLayout(new FlowLayout());
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        JLabel label1 = new JLabel("Enter First Number:");
        JTextField textField1 = new JTextField(10);

        JLabel label2 = new JLabel("Enter Second Number:");
        JTextField textField2 = new JTextField(10);

        JLabel resultLabel = new JLabel("Result will appear here");
        frame.add(label1);
        frame.add(textField1);

        frame.add(label2);
        frame.add(textField2);

        frame.add(resultLabel);

        frame.addMouseListener(new MouseAdapter() {
            public void mousePressed(MouseEvent e) {
                try {
                    int num1 = Integer.parseInt(textField1.getText());
```

```

        int num2 = Integer.parseInt(textField2.getText());
        int sum = num1 + num2;
        resultLabel.setText("Sum: " + sum);
    } catch (NumberFormatException ex) {
        resultLabel.setText("Invalid Input");
    }
}

public void mouseReleased(MouseEvent e) {
    try {
        int num1 = Integer.parseInt(textField1.getText());
        int num2 = Integer.parseInt(textField2.getText());
        int diff = num1 - num2;
        resultLabel.setText("Difference: " + diff);
    } catch (NumberFormatException ex) {
        resultLabel.setText("Invalid Input");
    }
}
});

frame.setVisible(true);
}
}

```

Output:

```

sakshamrimal@sakshamrimal-Latitude-5430:~/java/lab$ cd lab-4_Saksham\ /
sakshamrimal@sakshamrimal-Latitude-5430:~/java/lab/lab-4_Saksham $ cd src
sakshamrimal@sakshamrimal-Latitude-5430:~/java/lab/lab-4_Saksham /src$ javac Lab4SumDifference.java
sakshamrimal@sakshamrimal-Latitude-5430:~/java/lab/lab-4_Saksham /src$ java Lab4SumDifference

```

Sum and Difference

Enter First Number: 2 Enter Second Number: 3

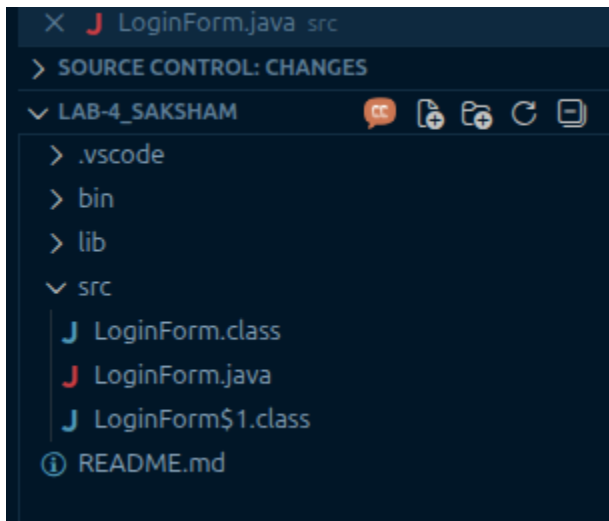
Difference: -1

Enter First Number: 1 Enter Second Number: 2

Sum: 3

LAB-5 : Create a Simple Login Form using GridLayout in Java

SOLUTION EXPLORER:



PROGRAM:

```
import javax.swing.*.*;
import java.awt.*.*;
import java.awt.event.*;

public class LoginFormGrid {

    public static void main(String[] args) {

        JFrame frame = new JFrame("Simple Login Form");
        frame.setSize(350, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        frame.setLayout(new GridLayout(4, 2, 10, 10));
        JLabel userLabel = new JLabel("Username:");
        JTextField userField = new JTextField();

        JLabel passLabel = new JLabel("Password:");
        JPasswordField passField = new JPasswordField();

        JButton loginButton = new JButton("Login");
        JButton resetButton = new JButton("Reset");
        frame.add(userLabel);
        frame.add(userField);

        frame.add(passLabel);
        frame.add(passField);

        frame.add(loginButton);
        frame.add(resetButton);

        loginButton.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
```

```

String username = userField.getText();
String password = new String(passField.getPassword());

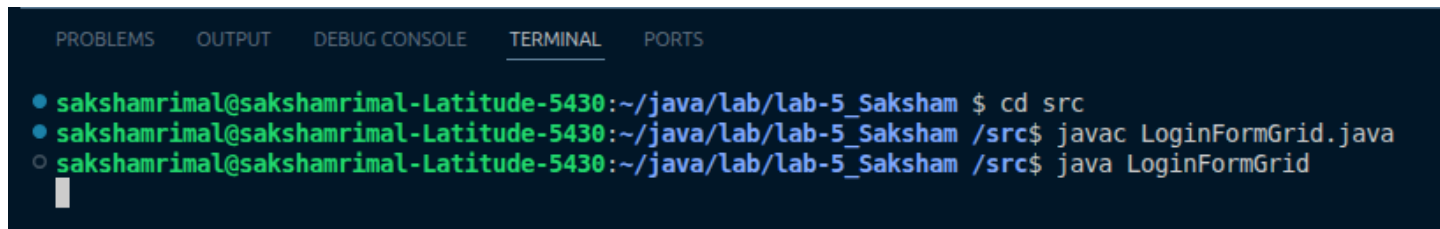
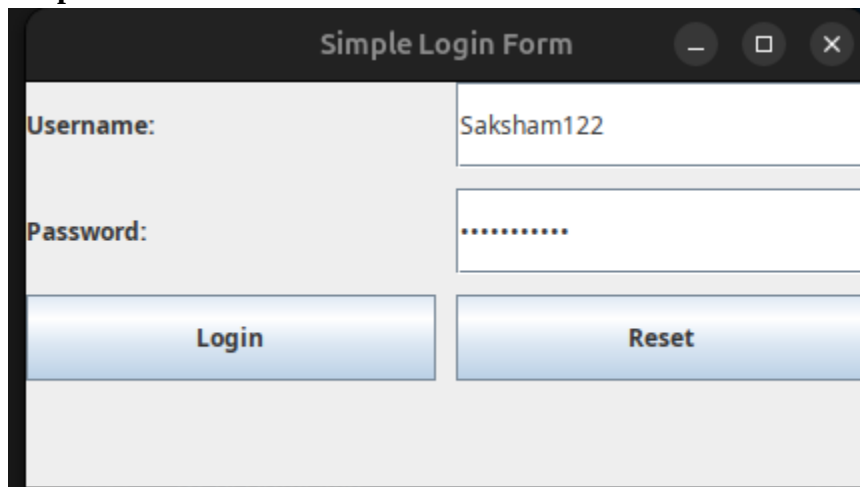
if (username.equals("admin") && password.equals("1234")) {
    JOptionPane.showMessageDialog(frame, "Login Successful!");
} else {
    JOptionPane.showMessageDialog(frame, "Invalid Credentials!");
}
}
});

// Reset Button Action
resetButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        userField.setText("");
        passField.setText("");
    }
});

frame.setVisible(true);
}
}

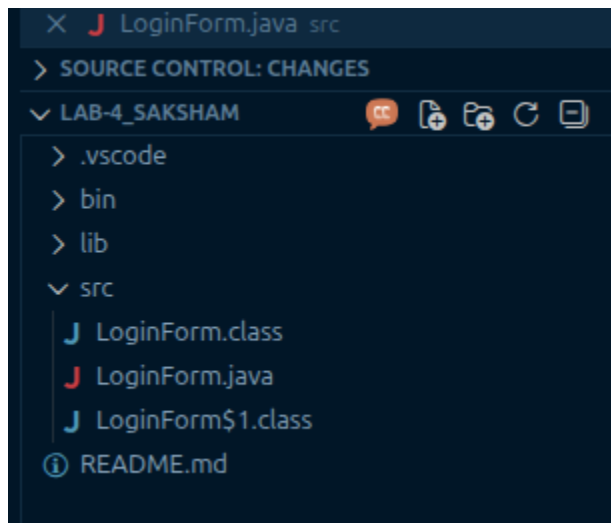
```

Output:



LAB-6 : Create a Simple Login Form using GridBagLayout in Java

SOLUTION EXPLORER:



PROGRAM:

```
import javax.swing.*.*;
import java.awt.*.*;
import java.awt.event.*;

public class LoginForm {

    public static void main(String[] args) {

        JFrame frame = new JFrame("Simple Login Form");
        frame.setSize(400, 250);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(new GridBagLayout());
        GridBagConstraints gbc = new GridBagConstraints();
        gbc.insets = new Insets(5, 5, 5, 5); // spacing
        JLabel userLabel = new JLabel("Username:");
        JTextField userField = new JTextField(15);
        JLabel passLabel = new JLabel("Password:");
        JPasswordField passField = new JPasswordField(15);
        JButton loginButton = new JButton("Login");
        gbc.gridx = 0;
        gbc.gridy = 0;
        frame.add(userLabel, gbc);
        gbc.gridx = 1;
        gbc.gridy = 0;
        frame.add(userField, gbc);
        gbc.gridx = 0;
        gbc.gridy = 1;
        frame.add(passLabel, gbc);
        gbc.gridx = 1;
        gbc.gridy = 1;
        frame.add(passField, gbc);
        gbc.gridx = 1;
```

```

gbc.gridy = 2;
gbc.anchor = GridBagConstraints.CENTER;
frame.add(loginButton, gbc);

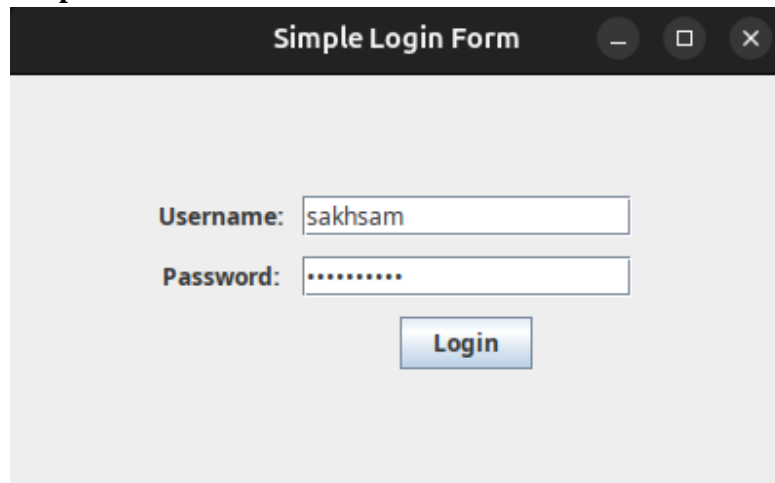
// Button Action
loginButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        String username = userField.getText();
        String password = new String(passField.getPassword());

        if (username.equals("admin") && password.equals("1234")) {
            JOptionPane.showMessageDialog(frame, "Login Successful!");
        } else {
            JOptionPane.showMessageDialog(frame, "Invalid Credentials!");
        }
    }
});

frame.setVisible(true);
}
}

```

Output:



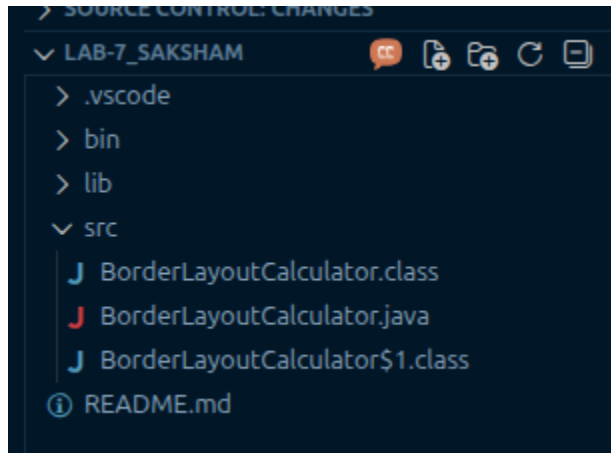
```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
java - src + v [icon] [icon] [icon] ... [icon] [icon] X
● sakshamrimal@sakshamrimal-Latitude-5430:~/java/lab/lab-4_Saksham$ cd src
● sakshamrimal@sakshamrimal-Latitude-5430:~/java/lab/lab-4_Saksham/src$ javac LoginForm.java
○ sakshamrimal@sakshamrimal-Latitude-5430:~/java/lab/lab-4_Saksham/src$ java LoginForm

```

LAB-7 : Write a program that divides the frame into five regions by using border layout and then add panels in the east, north and center region. Finally add some descriptive label in the north panel, buttons with icon in the east panel and a sample form in the center panel. You can further subdivide the center panel, if necessary. Prepare a program with three text boxes First Number, Second Number, and Result and four buttons add, subtract, multiply and divide. Handle the events to perform the required operation and display results.

SOLUTION EXPLORER:



PROGRAM:

```
import javax.swing.*.*;
import javax.swing.*.*;
import java.awt.*.*;
import java.awt.event.*;

public class BorderLayoutCalculator {

    public static void main(String[] args) {

        // Create main frame
        JFrame frame = new JFrame("BorderLayout Calculator");
        frame.setSize(600, 400);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(new BorderLayout(10, 10));

        // ===== North Panel =====
        JPanel northPanel = new JPanel();
        JLabel titleLabel = new JLabel("Simple Calculator Using BorderLayout", JLabel.CENTER);
        titleLabel.setFont(new Font("Arial", Font.BOLD, 18));
        northPanel.add(titleLabel);

        // Add north panel to frame
        frame.add(northPanel, BorderLayout.NORTH);

        // ===== East Panel =====
        JPanel eastPanel = new JPanel();
        eastPanel.setLayout(new GridLayout(4, 1, 5, 5));
```

```

JButton addIconButton = new JButton("Add", UIManager.getIcon("OptionPane.informationIcon"));
JButton subIconButton = new JButton("Subtract", UIManager.getIcon("OptionPane.warningIcon"));
JButton mulIconButton = new JButton("Multiply", UIManager.getIcon("OptionPane.errorIcon"));
JButton divIconButton = new JButton("Divide", UIManager.getIcon("OptionPane.questionIcon"));

eastPanel.add(addIconButton);
eastPanel.add(subIconButton);
eastPanel.add(mulIconButton);
eastPanel.add(divIconButton);

// Add east panel to frame
frame.add(eastPanel, BorderLayout.EAST);

// ===== Center Panel =====
JPanel centerPanel = new JPanel();
centerPanel.setLayout(new GridBagLayout());
GridBagConstraints gbc = new GridBagConstraints();
gbc.insets = new Insets(5, 5, 5, 5);
gbc.fill = GridBagConstraints.HORIZONTAL;

// Labels and TextFields
JLabel firstLabel = new JLabel("First Number:");
JTextField firstField = new JTextField(10);

JLabel secondLabel = new JLabel("Second Number:");
JTextField secondField = new JTextField(10);

JLabel resultLabel = new JLabel("Result:");
JTextField resultField = new JTextField(10);
resultField.setEditable(false);

// Buttons
JButton addButton = new JButton("Add");
JButton subButton = new JButton("Subtract");
JButton mulButton = new JButton("Multiply");
JButton divButton = new JButton("Divide");

// Place components in grid
gbc.gridx = 0;
gbc.gridy = 0;
centerPanel.add(firstLabel, gbc);

gbc.gridx = 1;
centerPanel.add(firstField, gbc);

gbc.gridx = 0;
gbc.gridy = 1;
centerPanel.add(secondLabel, gbc);

gbc.gridx = 1;
centerPanel.add(secondField, gbc);

```

```

gbc.gridx = 0;
gbc.gridy = 2;
centerPanel.add(resultLabel, gbc);

gbc.gridx = 1;
centerPanel.add(resultField, gbc);

gbc.gridx = 0;
gbc.gridy = 3;
centerPanel.add(addButton, gbc);

gbc.gridx = 1;
centerPanel.add(subButton, gbc);

gbc.gridx = 0;
gbc.gridy = 4;
centerPanel.add(mulButton, gbc);

gbc.gridx = 1;
centerPanel.add(divButton, gbc);

// Add center panel to frame
frame.add(centerPanel, BorderLayout.CENTER);

// ===== Event Handling =====
ActionListener calcAction = new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        try {
            double num1 = Double.parseDouble(firstField.getText());
            double num2 = Double.parseDouble(secondField.getText());
            double res = 0;

            if (e.getSource() == addButton || e.getSource() == addIconButton)
                res = num1 + num2;
            else if (e.getSource() == subButton || e.getSource() == subIconButton)
                res = num1 - num2;
            else if (e.getSource() == mulButton || e.getSource() == mulIconButton)
                res = num1 * num2;
            else if (e.getSource() == divButton || e.getSource() == divIconButton) {
                if (num2 == 0) {
                    resultField.setText("Cannot divide by zero");
                    return;
                }
                res = num1 / num2;
            }

            resultField.setText(String.valueOf(res));

        } catch (NumberFormatException ex) {
            resultField.setText("Invalid Input");
        }
    }
};

```

```

// Attach listeners
addButton.addActionListener(calcAction);
subButton.addActionListener(calcAction);
mulButton.addActionListener(calcAction);
divButton.addActionListener(calcAction);

addIconButton.addActionListener(calcAction);
subIconButton.addActionListener(calcAction);
mulIconButton.addActionListener(calcAction);
divIconButton.addActionListener(calcAction);

// Show frame
frame.setVisible(true);
}
}

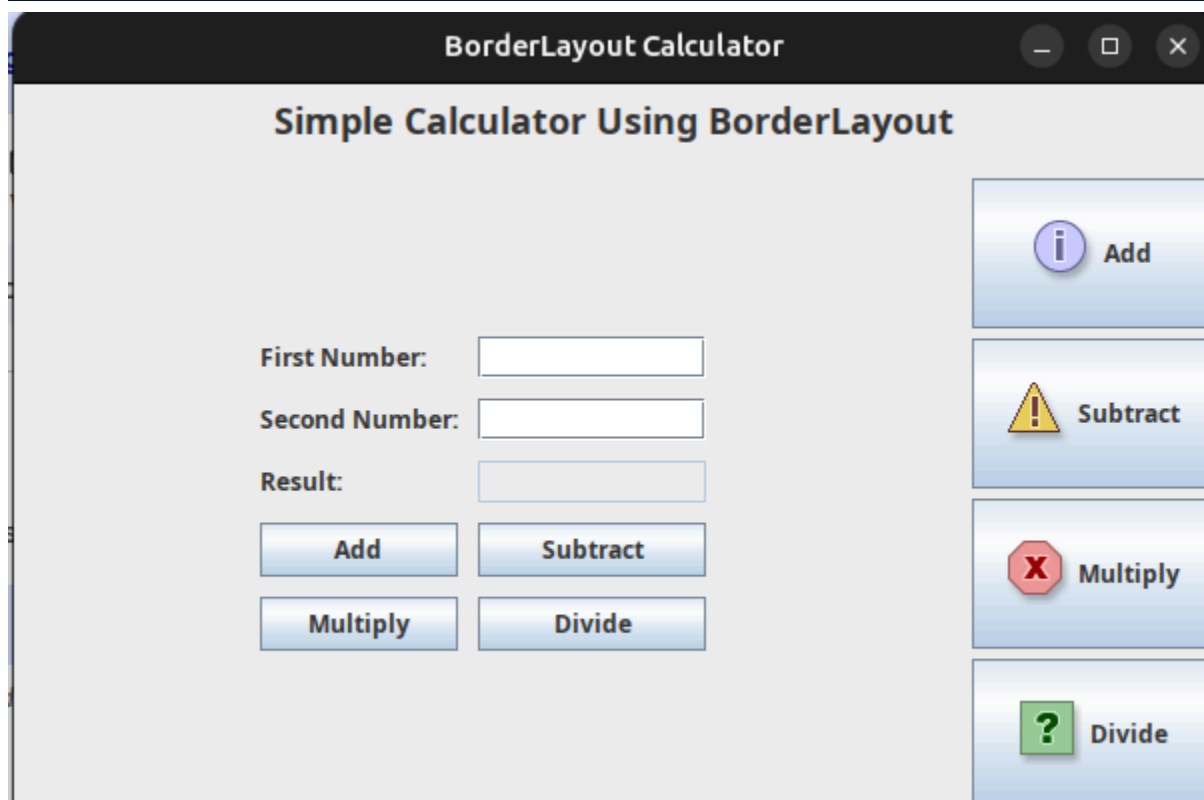
```

Output:

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
• sakshamrimal@sakshamrimal-Latitude-5430:~/java/lab/lab-7_Saksham/src$ javac BorderLayoutCalculator.java
○ sakshamrimal@sakshamrimal-Latitude-5430:~/java/lab/lab-7_Saksham/src$ java BorderLayoutCalculator

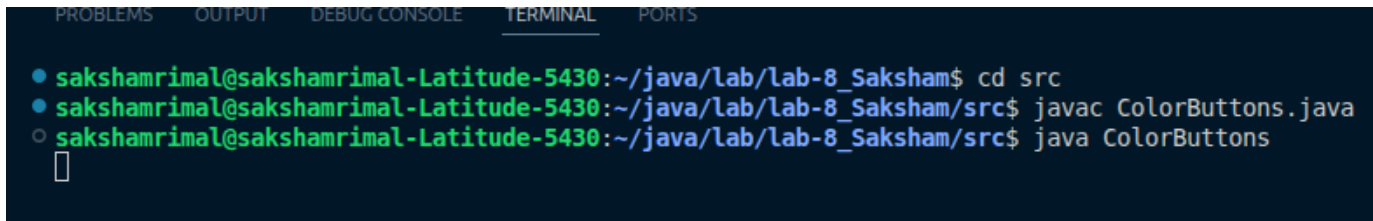
```



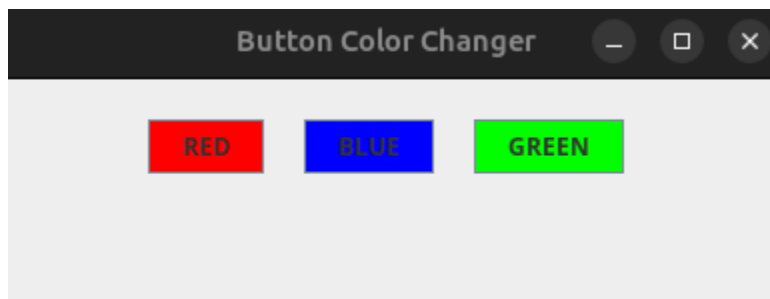

```
        break;
    case "BLUE":
        source.setBackground(Color.BLUE);
        break;
    case "GREEN":
        source.setBackground(Color.GREEN);
        break;
    }
}
};

redButton.addActionListener(colorAction);
blueButton.addActionListener(colorAction);
greenButton.addActionListener(colorAction);
frame.setVisible(true);
}
}
```

Output:

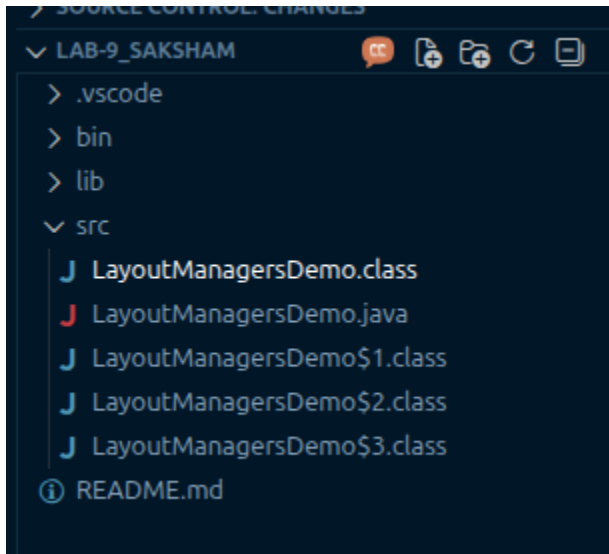


```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● sakshamrimal@sakshamrimal-Latitude-5430:~/java/lab/lab-8_Saksham$ cd src
● sakshamrimal@sakshamrimal-Latitude-5430:~/java/lab/lab-8_Saksham/src$ javac ColorButtons.java
○ sakshamrimal@sakshamrimal-Latitude-5430:~/java/lab/lab-8_Saksham/src$ java ColorButtons
█
```



LAB-9 : Write a Java program using Swing that demonstrates the use of different layout managers (FlowLayout, GridLayout, and BorderLayout).

SOLUTION EXPLORER:



PROGRAM:

```
import javax.swing.*.*;
import java.awt.*.*;
import java.awt.event.*;

public class LayoutManagersDemo {

    public static void main(String[] args) {

        JFrame frame = new JFrame("Layout Managers Demo");
        frame.setSize(500, 300);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(new BorderLayout(10, 10));
        JPanel northPanel = new JPanel(new FlowLayout(FlowLayout.CENTER, 20, 10));
        JLabel nameLabel = new JLabel("Name");
        JLabel rollLabel = new JLabel("Roll No");
        JLabel classLabel = new JLabel("Class");

        northPanel.add(nameLabel);
        northPanel.add(rollLabel);
        northPanel.add(classLabel);

        frame.add(northPanel, BorderLayout.NORTH);
        JPanel centerPanel = new JPanel(new GridLayout(3, 2, 10, 10));

        JLabel userLabel = new JLabel("Username:");
        JTextField userField = new JTextField();

        JLabel passLabel = new JLabel("Password:");
```

```

JPasswordField passField = new JPasswordField();

JCheckBox rememberCheck = new JCheckBox("Remember Me");
JButton loginButton = new JButton("Login");

centerPanel.add(userLabel);
centerPanel.add(userField);

centerPanel.add(passLabel);
centerPanel.add(passField);

centerPanel.add(rememberCheck);
centerPanel.add(loginButton);

frame.add(centerPanel, BorderLayout.CENTER);

JPanel southPanel = new JPanel(new FlowLayout(FlowLayout.CENTER, 20, 10));
JButton resetButton = new JButton("Reset");
JButton exitButton = new JButton("Exit");
southPanel.add(resetButton);
southPanel.add(exitButton);
frame.add(southPanel, BorderLayout.SOUTH);
loginButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        String username = userField.getText();
        JOptionPane.showMessageDialog(frame, "Logged in as: " + username);
    }
});
resetButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        userField.setText("");
        passField.setText("");
        rememberCheck.setSelected(false);
    }
});
exitButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        frame.dispose();
    }
});

frame.setVisible(true);
}
}

```

Output:

Layout Managers Demo

Name	Roll No	Class
------	---------	-------

Username:

Password:

☒ Remember Me

Login

Reset Exit

Layout Managers Demo

Name	Roll No	Class
------	---------	-------

Username:

Password:

☒ Remember Me

Login

Reset Exit

Message

Logged in as: saksham07

OK

Layout Managers Demo

Name	Roll No	Class
------	---------	-------

Username:

Password:

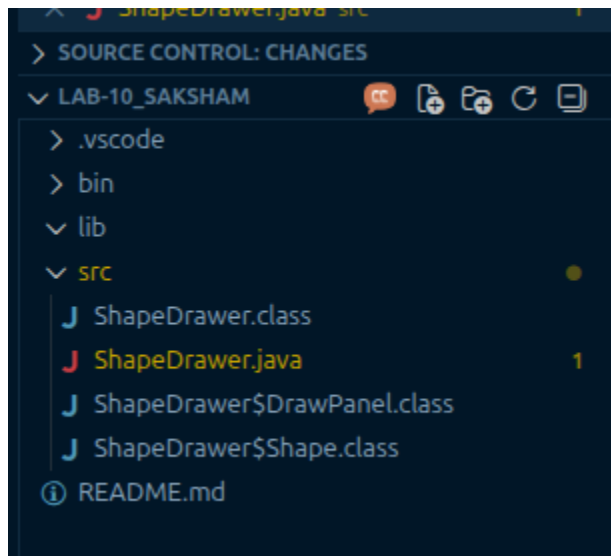
☐ Remember Me

Login

Reset Exit

LAB-10 : Design a form with three buttons with captions “Circle,” “Square,” and “Triangle.” , When the user clicks the “Circle” button, draw a red circle in the center of the frame. When the user clicks the “Square” button, draw a blue square in the center of the frame, When the user clicks the “Triangle” button, draw a green triangle in the center of the frame

SOLUTION EXPLORER:



PROGRAM:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class ShapeDrawer {

    enum Shape { NONE, CIRCLE, SQUARE, TRIANGLE }

    public static void main(String[] args) {
        JFrame frame = new JFrame("Shape Drawer");
        frame.setSize(500, 500);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(new BorderLayout());
        JPanel buttonPanel = new JPanel(new FlowLayout());
        JButton circleButton = new JButton("Circle");
        JButton squareButton = new JButton("Square");
        JButton triangleButton = new JButton("Triangle");
        buttonPanel.add(circleButton);
        buttonPanel.add(squareButton);
        buttonPanel.add(triangleButton);
        frame.add(buttonPanel, BorderLayout.SOUTH);
        DrawPanel drawPanel = new DrawPanel();
        frame.add(drawPanel, BorderLayout.CENTER);
        circleButton.addActionListener(e -> {
            drawPanel.setShape(Shape.CIRCLE);
        });
    }
}
```

```

squareButton.addActionListener(e -> {
    drawPanel.setShape(Shape.SQUARE);
});
triangleButton.addActionListener(e -> {
    drawPanel.setShape(Shape.TRIANGLE);
});
frame.setVisible(true);
}

static class DrawPanel extends JPanel {
    private Shape currentShape = Shape.NONE;

    public void setShape(Shape shape) {
        currentShape = shape;
        repaint(); // Repaint panel to show new shape
    }
    @Override
    protected void paintComponent(Graphics g) {
        super.paintComponent(g);
        int width = getWidth();
        int height = getHeight();

        int size = 100; // Size of the shape
        int x = (width - size) / 2;
        int y = (height - size) / 2;

        switch (currentShape) {
            case CIRCLE:
                g.setColor(Color.RED);
                g.fillOval(x, y, size, size);
                break;

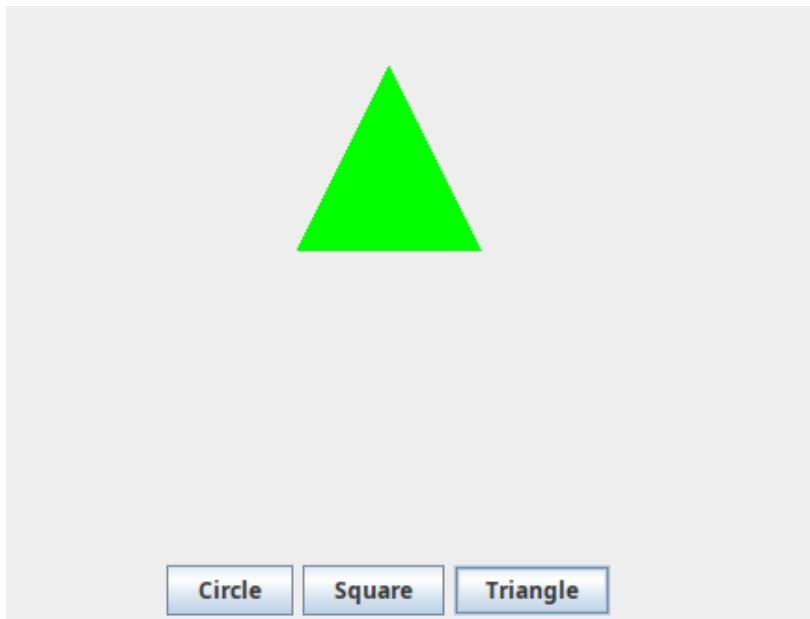
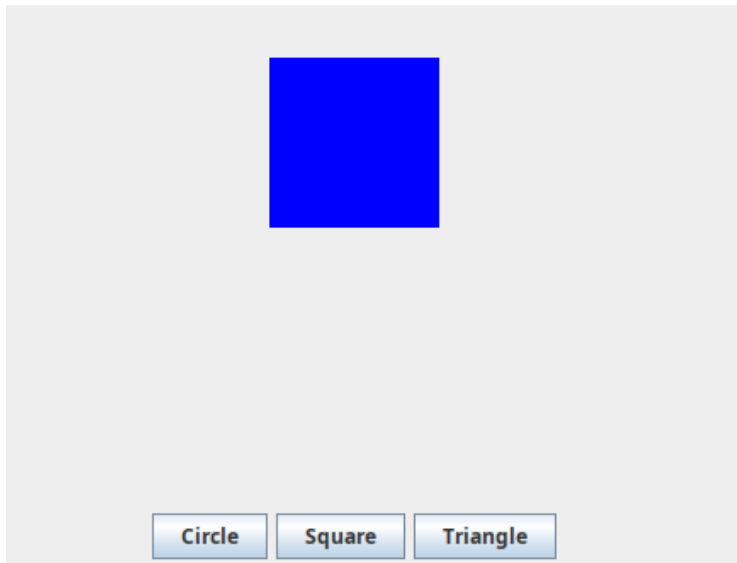
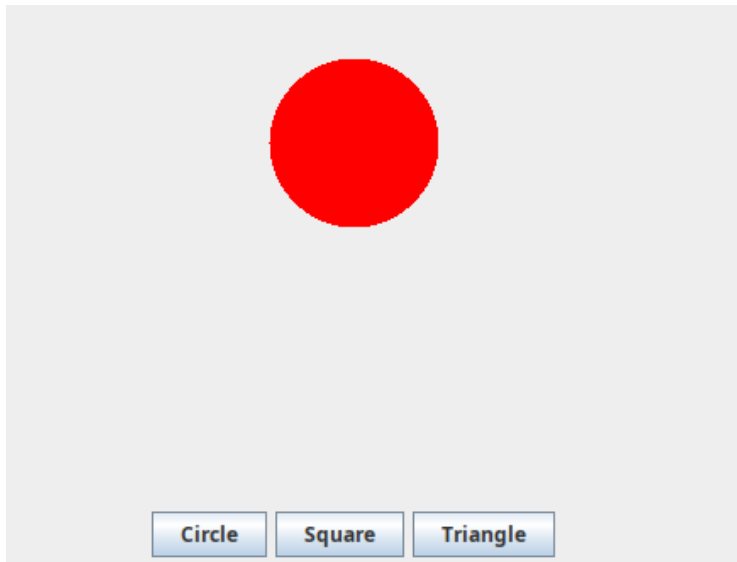
            case SQUARE:
                g.setColor(Color.BLUE);
                g.fillRect(x, y, size, size);
                break;

            case TRIANGLE:
                g.setColor(Color.GREEN);
                int[] xPoints = { x + size / 2, x, x + size };
                int[] yPoints = { y, y + size, y + size };
                g.fillPolygon(xPoints, yPoints, 3);
                break;

            case NONE:
                // Do nothing
                break;
        }
    }
}

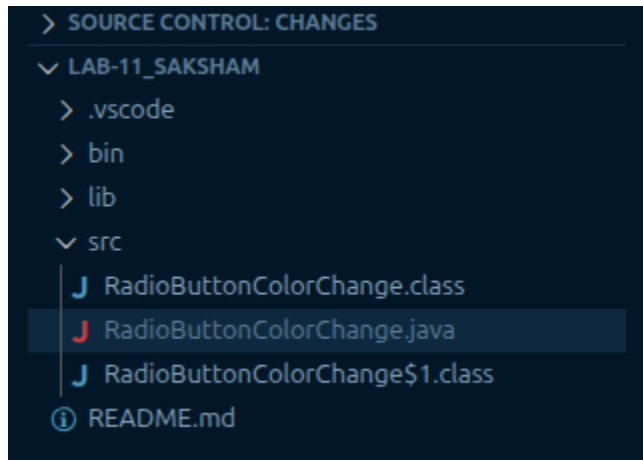
```

Output:



LAB-11 : Design a Swing application that contains three radio buttons labeled "Red", "Green", and "Blue". When the user selects a radio button, the background color of the window should change to the selected color. Implement event handling using an ItemListener.

SOLUTION EXPLORER:



PROGRAM:

```
import javax.swing.*.*;
import javax.swing.*.*;
import java.awt.*.*;
import java.awt.event.*;

public class RadioButtonColorChange {
    public static void main(String[] args) {

        JFrame frame = new JFrame("Radio Button Color Changer");
        frame.setSize(400, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(new FlowLayout());

        JRadioButton redButton = new JRadioButton("Red");
        JRadioButton greenButton = new JRadioButton("Green");
        JRadioButton blueButton = new JRadioButton("Blue");

        ButtonGroup group = new ButtonGroup();
        group.add(redButton);
        group.add(greenButton);
        group.add(blueButton);

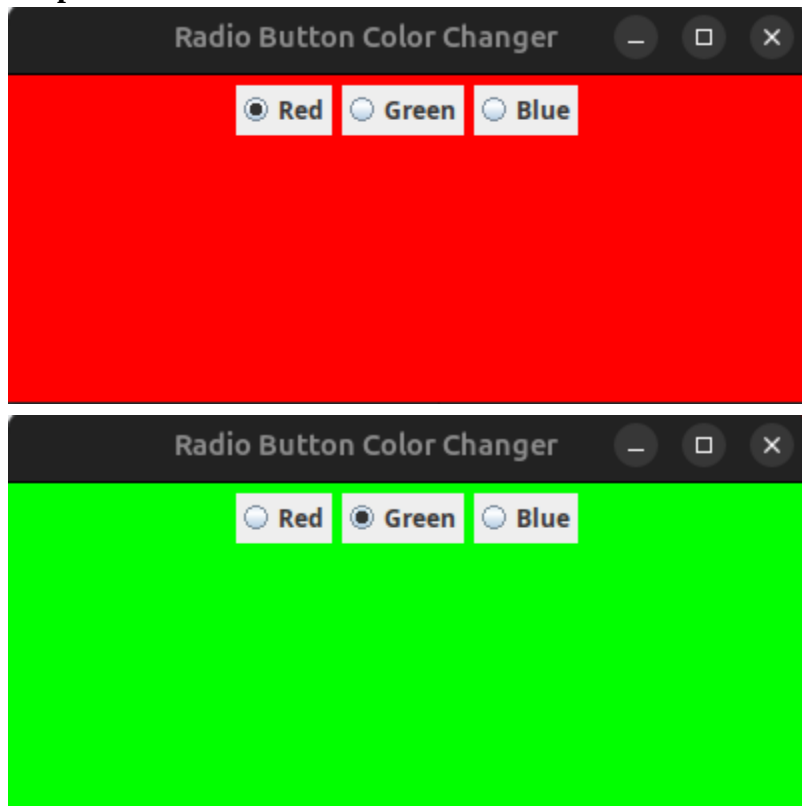
        frame.add(redButton);
        frame.add(greenButton);
        frame.add(blueButton);

        ItemListener colorListener = new ItemListener() {
            @Override
            public void itemStateChanged(ItemEvent e) {
                JRadioButton source = (JRadioButton) e.getSource();
```

```
if (e.getStateChange() == ItemEvent.SELECTED) {
    switch (source.getText()) {
        case "Red":
            frame.getContentPane().setBackground(Color.RED);
            break;
        case "Green":
            frame.getContentPane().setBackground(Color.GREEN);
            break;
        case "Blue":
            frame.getContentPane().setBackground(Color.BLUE);
            break;
    }
}
};
redButton.addItemListener(colorListener);
greenButton.addItemListener(colorListener);
blueButton.addItemListener(colorListener);

frame.getContentPane().setBackground(Color.LIGHT_GRAY);
frame.setVisible(true);
}
```

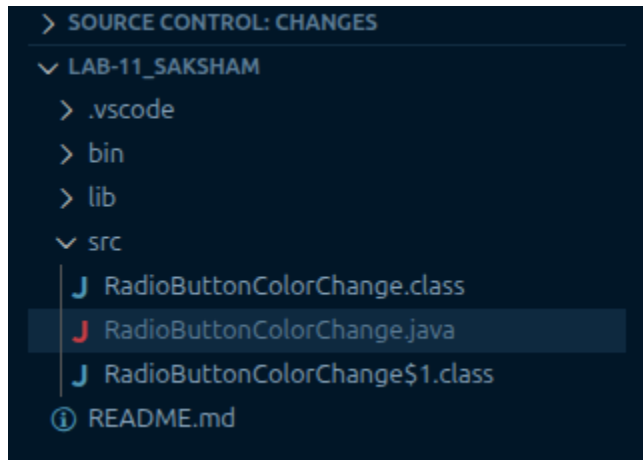
Output:





LAB-12 : Write a Java Swing program to create a toolbar with three buttons. Each button should display an icon (New, Login, and Logout). Place the toolbar at the top of the window using a suitable layout. When a button is clicked, display a message dialog showing the name of the action (e.g., “New File Created”, “Login Successful”, “Logged Out”).

SOLUTION EXPLORER:



PROGRAM:

```
import javax.swing.*.*;
import javax.swing.*.*;
import java.awt.*.*;
import java.awt.event.*;
public class ToolbarDemo {

    public static void main(String[] args) {

        JFrame frame = new JFrame("Toolbar Demo");
        frame.setSize(500, 200);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(new BorderLayout());
        JToolBar toolBar = new JToolBar();
        JButton newButton = new JButton("New");
        JButton loginButton = new JButton("Login");
        JButton logoutButton = new JButton("Logout");
        newButton.setIcon(UIManager.getIcon("FileView.fileIcon"));
        loginButton.setIcon(UIManager.getIcon("OptionPane.informationIcon"));
        logoutButton.setIcon(UIManager.getIcon("OptionPane.errorIcon"));
        toolBar.add(newButton);
        toolBar.add(loginButton);
        toolBar.add(logoutButton);

        frame.add(toolBar, BorderLayout.NORTH);
        newButton.addActionListener(e ->
            JOptionPane.showMessageDialog(frame, "New File Created")
        );

        loginButton.addActionListener(e ->
            JOptionPane.showMessageDialog(frame, "Login Successful")
        );
    }
}
```

```

    );

    logoutButton.addActionListener(e ->
        JOptionPane.showMessageDialog(frame, "Logged Out")
    );

    frame.setVisible(true);
}
}

```

Output:

